

Chapter 2 Exercises

Directions: For exercises 1 - 8, determine whether the equation of the curve can be written as a linear function.

1. $y = \frac{1}{4}x + 6$

2. $y = 3x - 5$

3. $y = 3x^2 - 2$

4. $3x + 5y = 15$

5. $3x^2 + 5y = 15$

6. $3x + 5y^2 = 15$

7. $-2x^2 + 3y^2 = 6$

8. $-\frac{x-3}{5} = 2y$

Directions: For exercises 9 - 18, determine whether each function is increasing or decreasing.

9. $f(x) = 4x + 3$

10. $g(x) = 5x + 6$

11. $a(x) = 5 - 2x$

12. $b(x) = 8 - 3x$

13. $h(x) = -2x + 4$

14. $k(x) = -4x + 1$

15. $j(x) = \frac{1}{2}x - 3$

16. $p(x) = \frac{1}{4}x - 5$

17. $n(x) = -\frac{1}{3}x - 2$

18. $m(x) = -\frac{3}{8}x + 3$

Directions: For exercises 19 - 23, find the slope of the line that passes through the two given points.

19. (2, 4) and (4, 10)

20. (1, 5) and (4, 11)

21. (-1, 4) and (5, 2)

22. (8, -2) and (4, 6)

23. (6, 11) and (-4, 3)

Directions: For exercises 24 - 31, given each set of information, find a linear equation satisfying the conditions.

24. $f(-5) = -4$, and $f(5) = 2$

25. $f(-1) = 4$ and $f(5) = 1$

26. $(2,4)$ and $(4,10)$

27. Passes through $(1,5)$ and $(4,11)$

28. Passes through $(-1, 4)$ and $(5, 2)$

29. Passes through $(-2, 8)$ and $(4, 6)$

30. x -intercept at $(-2, 0)$ and y -intercept at $(0, -3)$

31. x -intercept at $(-5, 0)$ and y -intercept at $(0,4)$

Directions: For exercises 32 - 37, determine whether the lines given by the equations below are parallel, perpendicular, or neither.

$$4x - 7y = 10$$

32. $7x + 4y = 1$

$$3y + x = 12$$

33. $-y = 8x + 1$

$$3y + 4x = 12$$

34. $-6y = 8x + 1$

$$6x - 9y = 10$$

35. $3x + 2y = 1$

$$y = \frac{2}{3}x + 1$$

36. $3x + 2y = 1$

$$y = \frac{3}{4}x + 1$$

37. $-3x + 4y = 1$

Directions: For exercises 38 - 43, find the x - and y -intercepts of each equation.

38. $f(x) = -x + 2$

39. $g(x) = 2x + 4$

40. $h(x) = 3x - 5$

41. $k(x) = -5x + 1$

42. $-2x + 5y = 20$

43. $7x + 2y = 56$

Directions: For exercises 44 - 49, use the descriptions of each pair of lines given below to find the slopes of Line 1 and Line 2. Is each pair of lines parallel, perpendicular, or neither?

44. Line 1: Passes through (0,6) and (3, -24). Line 2: Passes through (-1,19) and (8, -71)

45. Line 1: Passes through (-8, -55) and (10, 89). Line 2: Passes through (9, -44) and (4, -14)

46. Line 1: Passes through (2,3) and (4, -1). Line 2: Passes through (6,3) and (8,5)

47. Line 1: Passes through (1,7) and (5,5). Line 2: Passes through (-1, -3) and (1,1)

48. Line 1: Passes through (0,5) and (3,3). Line 2: Passes through (1, -5) and (3, -2)

49. Line 1: Passes through (2,5) and (5, -1). Line 2: Passes through (-3,7) and (3, -5)

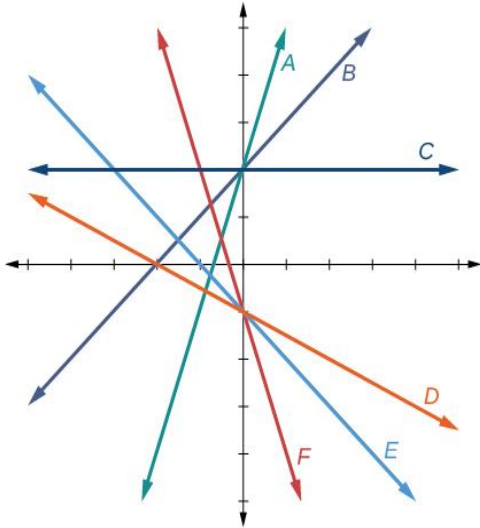
50. Write an equation for a line parallel to $f(x) = -5x - 3$ and passing through the point (2, -12).

51. Write an equation for a line parallel to $g(x) = 3x - 1$ and passing through the point (4,9).

52. Write an equation for a line perpendicular to $h(t) = -2t + 4$ and passing through the point (-4, -1).

53. Write an equation for a line perpendicular to $p(t) = 3t + 4$ and passing through the point (3,1).

Directions: For exercises 54 - 59, match the given linear equation with its graph.



54. $f(x) = -x - 1$

55. $f(x) = -2x - 1$

56. $f(x) = -\frac{1}{2}x - 1$

57. $f(x) = 2$

58. $f(x) = 2 + x$

59. $f(x) = 3x + 2$

Directions: For exercises 60 - 74, sketch the graph of each equation.

60. $f(x) = -2x - 1$

61. $g(x) = -3x + 2$

62. $h(x) = \frac{1}{3}x + 2$

63. $k(x) = \frac{2}{3}x - 3$

64. $f(t) = 3 + 2t$

65. $p(t) = -2 + 3t$

66. $x = 3$

67. $x = -2$

68. $r(x) = 4$

69. $q(x) = 3$

70. $4x = -9y + 36$

71. $\frac{x}{3} - \frac{y}{4} = 1$

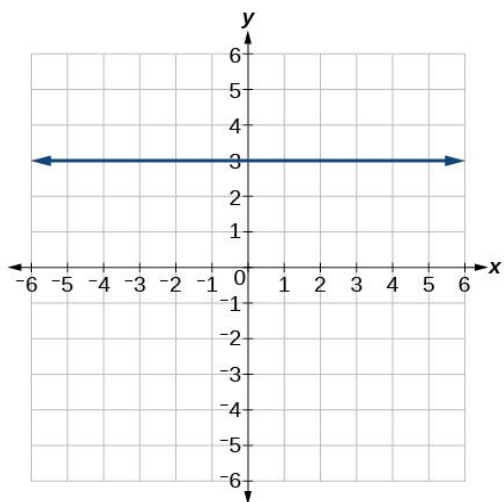
72. $3x - 5y = 15$

73. $3x = 15$

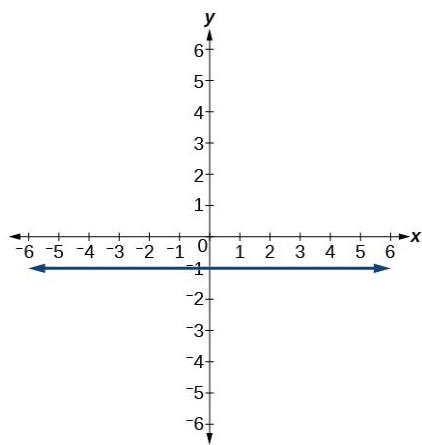
74. $3y = 12$

For exercises 75 – 78, write the equation of the line shown in the graph.

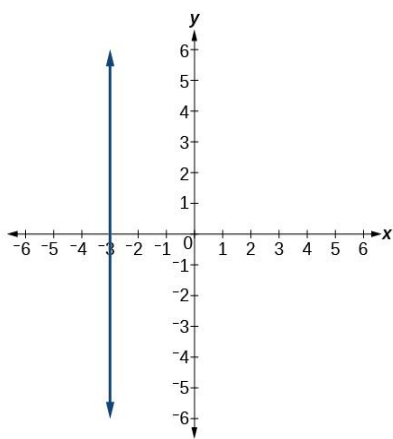
75.



76.



77.



78.

